

FUNCTIONAL DESCRIPTION

CONTAINER SERIES

CHP

For Type Series 400



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Subject to technical alterations in the interest of further development.

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1 General Description

The container unit is made up of the following main groups:

- Steel construction container enclosure
- MTU Series 4000 (gas engine with generator mounted onto base frame)
- MMC (for system control, control, diagnosis)
- MIP (MTU Interface Panel) for engine control, regulating, and diagnosis
- Generator power control panel
- Gas system (incl. controlled gas system)
- Gas compressor (optional)
- Ventilation system
- Cooling system for engine cooling water and mixture cooling
- Heat extraction HP circuit (optional)
- Exhaust gas heat extraction (optional)
- Exhaust system
- Lube oil system

NOTICE



The delivery object of MTU Onsite Energy satisfies the requirement specified in the EC Machinery Directive and in the German directives and standards.

SH-H-150

NOTICE



When using the unit supplied in a foreign country or special regions, MTU Onsite Energy is not responsible for adherence to the legal or other regulations / standards at the place of use.

SH-H-151

NOTICE



The images may not be identical to the actual design since some available customer options may be hidden or omitted on purpose. Furthermore, any tailor-made designs are not shown in these images.

SH-H-150

NOTICE

Please refer to wire diagram or operating instructions.

SH-H-150

NOTICE

Refer to operating instructions

SH-H-150

INFORMATION

Compliance with the currently valid specifications for fluids and lubricants is mandatory!

SH-H-023

2 Container

This container is a self-contained unit that comprises all the components necessary for the safe operation of the genset contained therein. The container unit has been pre-tested to the extent possible and is ready for operation following the installation of roof attachments and connection of onsite media (fuel gas, power, external supply for auxiliary drives, use of heat provided by the customer (if applicable)). The required tasks to be performed are limited to installing the roof attachments (if applicable) and connecting interfaces as described above, as well as those operating fluids available (cooling water, lubricants, etc.). Onsite commissioning tasks may vary depending on the design and scope of the system.

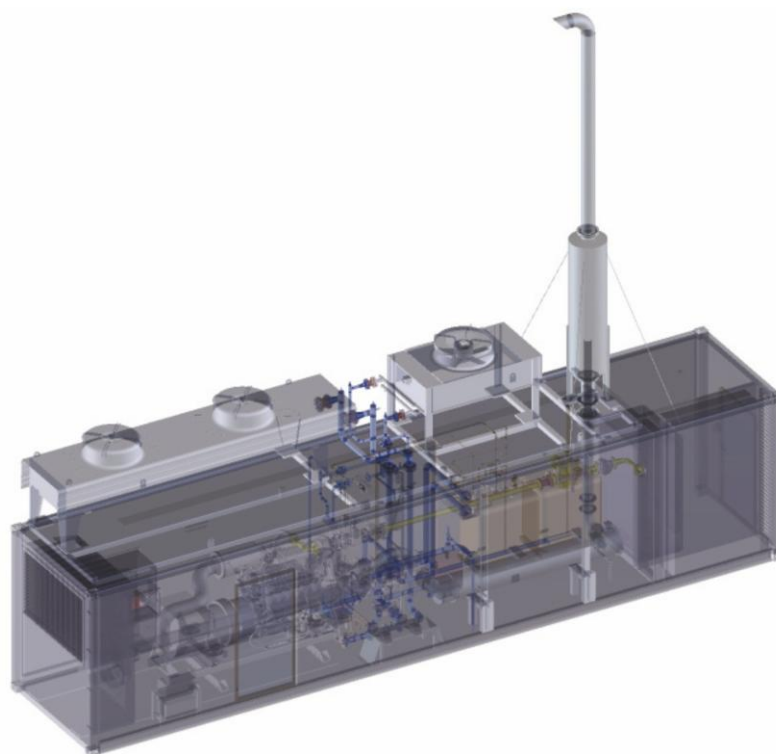
The container enclosure provides the necessary protection against the effects of weather. The container enclosure ensures that the noise created by the gas engine is kept to a defined minimum.

Containers are equipped with access and maintenance doors and removable structural elements, which enable components to be moved in and out of the container.

In order to protect the environment, the floor plates of the containers are designed as oil sumps and equipped with detection devices (holding capacity: 110% of the fluids inside the container). This is compliant with the *WHG* (German Federal Water Act).

The exact technical design of containers may vary, depending on the series and application.

2.1 CHP container series 400



GX-N-xx000028

2.1.1 Container enclosure

A modified version of the ISO 40ft High Cube Container, adapted to the special application, serves as base structure.

The container is stackable and easy to transport (CSC certified).

The container is typically divided into two rooms:

the machine room, which contains the genset, heat extraction system, and the auxiliary drive units, and a second, separate room for the switchgear.

The genset sits on support rails that are anchored into the floor of the machine room. The machine room accommodates further components necessary for the operation of the genset, e.g., gas, lubrication oil, or cooling systems, and the generator power control panel.

The control system for the facility (MTU Module Control, MMC) and the electrical components for the auxiliary drive units are located in the second, separate room.

The heat exchangers required for cooling the engine and the mixture, as well as the exhaust gas silencer are arranged on the roof. Following transport, these roof attachments should be installed on the roof of the container.

Onsite connections for gas, power supply, and auxiliary drive units are located along the side wall.

Onsite connections for using the heat (optional) of the engine cooling water can be found on the roof of the container.

2.2 Generator Output Field

The generator's power control panel is completely wired and mounted onto the side wall of the container.

It comprises the following main components

- Generator circuit breaker
- Connections provided by the customer
- Separate supply system for auxiliary drive units (optional)
- Terminals and interfaces for customer communication

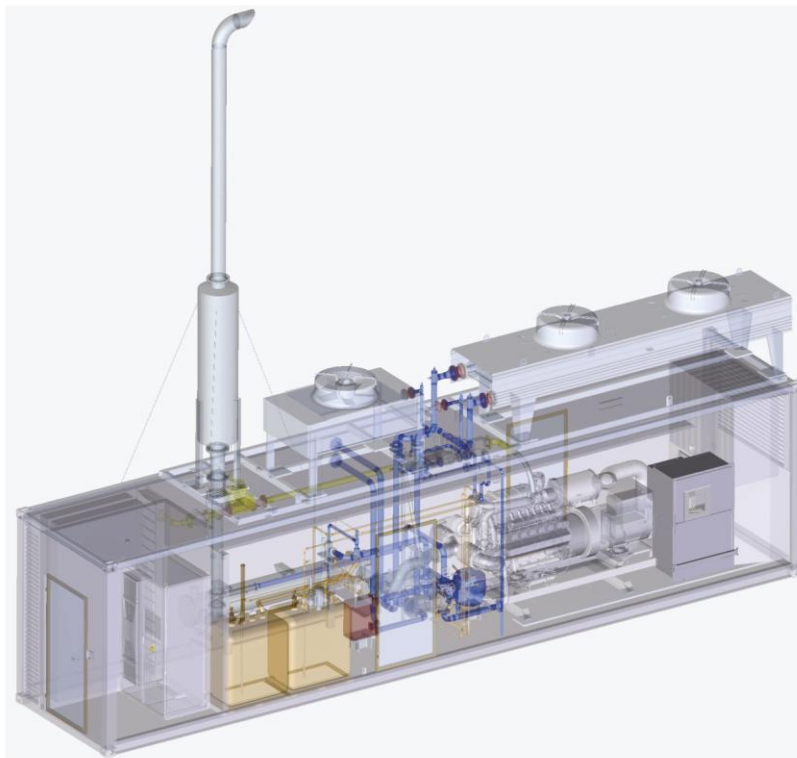
The generator power control panel is the interface to the customer's network.

This module is also used when disconnecting the system from the main power supply.

A separate feed point is used for providing the external feed (optional) required for island mode for the most important auxiliary drive units, which are required for safe operation.

The aforementioned components can be accessed and operated from the outside.

The place of installation is along the side wall of the container.



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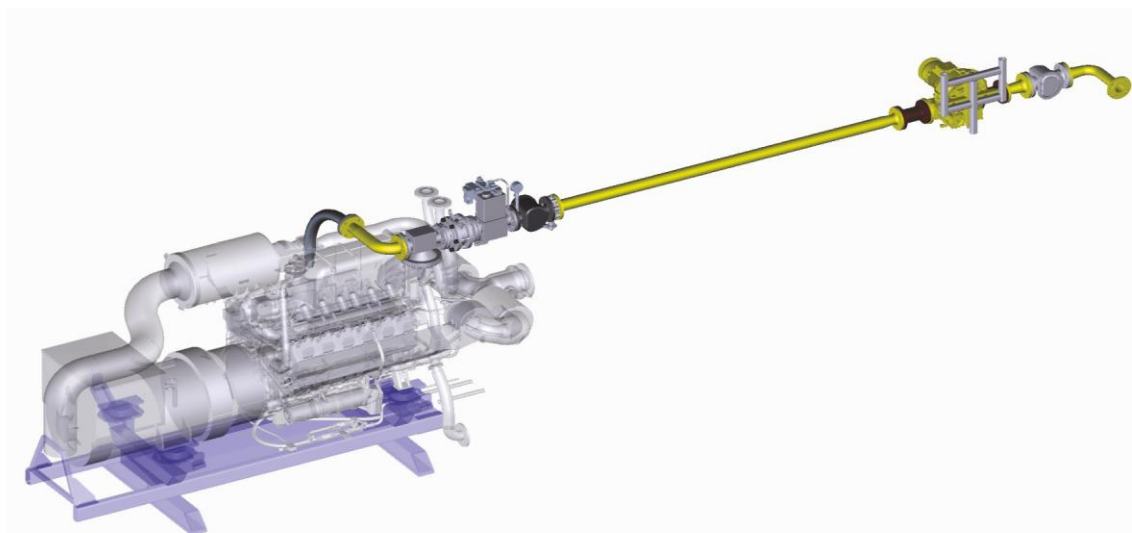
2.3 Gas System

Fully assembly ready – gas control system

It comprises the following components:

- Gas filter
- Two solenoid valves (or a double solenoid valve)
- Zero pressure regulator
- Valve tightness check
- Lambda control valve
- Flexible stainless steel hose assembly
- Shut-off devices
- SSV (safety shut-off valve) (included)
to be installed into gas line outside of the container.
(only for natural gas applications)
- Gas compressor (optional)
- Deflagration safeguard (not for natural gas applications)

A welded DIN-compliant flange in the container wall is used to provide an interface to the customer.



GX-N-xxx00023

NOTICE



Please refer to operating instructions and manufacturer's documentation.

SH-H-150

2.4 Ventilation system

A ventilation system is integrated into the container and ensures the following features.

- It ensures the proper amount of compressed air is supplied to the machine base (cp. safety concept).
- Transporting the engine's combustion air.
- Transporting the cooling air for the generator and other components installed in the system.

One or several fans are used to carry the necessary amount the air.

The operating mode of the fans as well as their activation may be indifferent, depending on the series of the container.

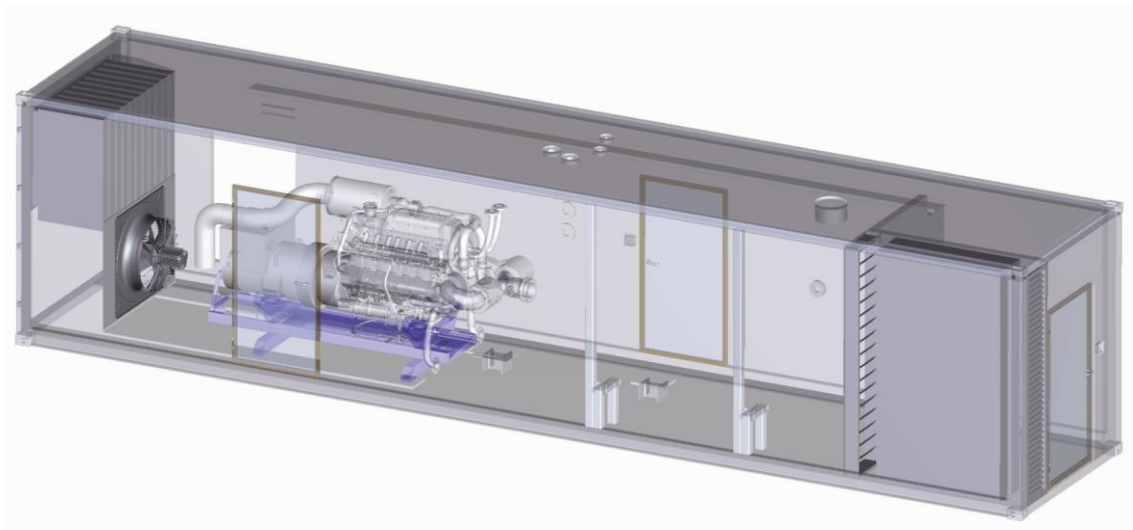
Powercontainer	Exhaust air fan
CHP Container	Supply air fan

The supply as well as the exhaust air ducts:

It comprises the following components:

- Sound dampeners
- Weather protection and screens to keep out small animals
- Air filter (optional)
- Motor-driven louvers

Outlet	Exhaust air via front side
Inlet	Supply air via front side



GX-N-xxx000024